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DPR Lecture: Neurologic Exam 2: Motor System, Coordination, Gait, Reflexes and Special Techniques

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Session Objectives

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- Review the screening Neurological Examination
- Learn how to perform a neurological assessment of the motor system, coordination, reflexes and gait including the use of special techniques
- Learn how to document the physical findings for a neurological examination

Source: Course Syllabus

Neurological Exam

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Start with the patient's history

Common or Concerning Symptoms:

- Headache
- Dizziness/Vertigo
- Weakness
- Numbness, Abnormal or Absent Sensation
- Syncope/ Near Syncope
- Seizures
- Tremors/ Involuntary Movements

Components of a Screening Neurological Examination

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- **Mental Status**—level of alertness, appropriateness of responses, orientation to date and place
- **Cranial Nerves**
- Visual Acuity
- Pupillary light reflex
- Eye movements
- Hearing
- Facial strength- smile, eye closure
- Speech
- **Motor System**
- Strength—shoulder abduction, elbow flexion/extension, wrist extension, finger abduction, hip flexion, knee flexion/extension, ankle dorsiflexion
- **Gait**—casual and tandem
- **Coordination**—fine finger movements, finger-to-nose or finger to chin
- **Reflexes**- Deep Tendon Reflexes and Plantar responses
- **Sensory System**—one modality at toes—can be light touch, pain/temperature, or proprioception

Bates, 13th Edition, Chapter 24- American Academy of Neurology Guidelines for a Screening Neurologic Examination

Motor System

- Motor pathways involve the upper motor neurons that will synapse with the lower motor neurons, then to the peripheral nerve structures and terminate at the neuromuscular junction.

Upper and Lower Motor Neurons

- **Upper motor neuron (UMN)** cell bodies are located in the motor strip of the cerebral cortex and several brainstem nuclei. Their axons synapse with the motor nuclei in the brainstem for the CNs and in the spinal cord for peripheral nerves
- **Lower motor neuron (LMN)** cell bodies (anterior horn cells) are in the spinal cord. Their axons transmit impulses to the peripheral nerves through the anterior roots and spinal nerves

Motor Function

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Principal Motor Pathways

- Corticospinal (pyramidal) tract
- Cerebellar System
- Basal Ganglion System

Principal Motor Pathways

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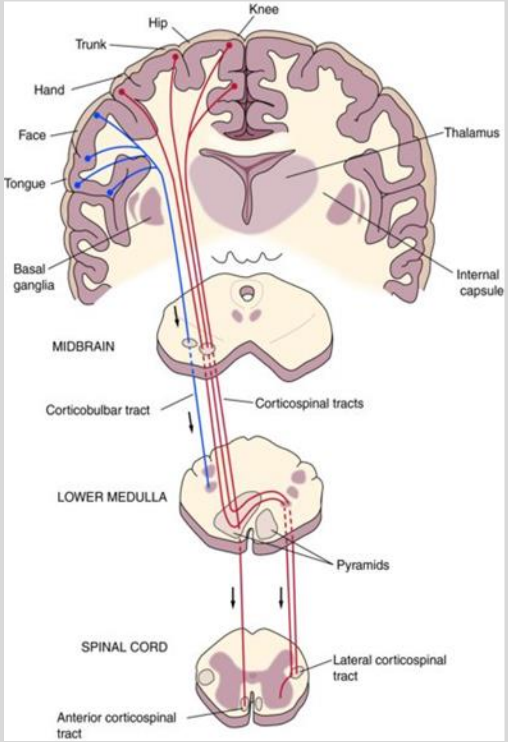
- **Corticospinal (pyramidal) system-** originates in the motor cortex, mediates voluntary movement and integrates skilled complicated movements through the stimulation and inhibition of certain muscle actions

  Consists of both upper and lower motor neurons

  Synapse on the LMN in the spinal cord that mediate movement

  Most fibers will cross over to contralateral side at the level of the medulla and spinal cord

  (Tracts that synapse in the brainstem with the cranial nerve motor nuclei are the **corticobulbar tracts**.)



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Principal Motor Pathways (continue)

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- **Basal Ganglion system-** help maintain muscle tone and control body movements (walking)
- **Cerebellar system-** receives sensory and motor input and coordinates motor activity- maintains equilibrium and assists to control posture

Upper/Lower Motor Neuron Lesions

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Upper Motor Neuron (UMN) lesions

- If UMN system has damage that occurs above the cross over in the medulla- motor impairment (weakness and paralysis) is on contralateral side
- If UMN system damage occurs below the cross over in the medulla- motor impairment is on the ipsilateral side
- Spasticity
- Muscle tone is increased
- Deep Tendon Reflexes (DTR) are exaggerated- hyperreflexia
- Babinski sign present (or Plantar dorsiflexion)

Lower Motor Neuron (LMN) lesions

- LMN system damage will have motor impairment (weakness and paralysis) on the ipsilateral side
- Muscle tone is decreased
- Deep Tendon Reflexes (DTR) are decreased (hyporeflexia) or absent
- Atrophy
- Fasciculations

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Lesions to the Corticospinal tract, Basal Ganglion and Cerebellar systems

- Corticospinal tract – weakness
- Basal Ganglia system - changes in muscle tone (mostly increase), disturbances in posture and gait, slowness or lack of spontaneous and automatic movements and various involuntary movements
- Cerebellar system – Impairment of coordination, gait and equilibrium and decrease muscle tone

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Motor System- Muscles

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Assessment

- Body Position- at rest and during movement
- Involuntary Movements- tremors, tics and fasciculations
- Muscle Bulk- contour and size of muscles, atrophy or wasting- unilateral or bilateral
- Muscle Tone- assess the muscle's resistance to passive stretch
- Muscle Strength- assess by asking the patient to actively resist your movement

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Muscle Atrophy

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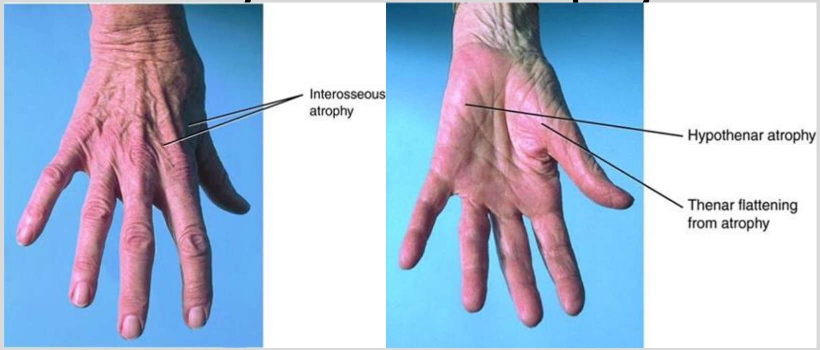
- 44 yo- no atrophy

84 yo- with atrophy



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Motor System

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Muscle Weakness

Proximal

- In Shoulder and /or hip girdle
- Identify by asking about difficulty combing hair
- Getting out of a chair, climbing stairs

Distal

- In the hands and /or feet
- Identify by asking about hand strength when opening a jar or tripp ing when walking

Muscle Strength Abnormalities

- Plegia or Paralysis- absent strength
- Paresis- strength is impaired
- Hemiparesis- weakness to one side of the body
- Hemiplegia- paralysis to one side of the body
- Monoparesis- weakness in an extremity
- Paraparesis- weakness of the legs
- Paraplegia- paralysis of the legs
- Quadriplegia- paralysis of all four limbs

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Muscle Strength Scale

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Scale for muscle strength

0—No muscular contraction detected

1—A barely detectable flicker or trace of contraction

2—Active movement of the body part with gravity eliminated

3—Active movement against gravity

4—Active movement against gravity and some resistance

5—Active movement against full resistance without evident fatigue. (normal strength)

Source: Medical Research Council. Aids to the examination of the peripheral nervous system. London: Bailliere Tindall, 1986.

Documentation of Muscle Strength- 1/5, 2/5, 3/5, 4/5, 5/5

Bates 12th Edition Chapter 17

Upper Extremities- MS- Flexion/Extension

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Flexion

Spinal nerve root innervations: C5, C6

- Patient is instructed to flex their arms at the elbow
- Examiner will instruct the patient to pull towards themselves (flex) as the examiner provides resistance

Extension

Spinal root innervations: C6, C7, C8

- Patient is instructed to flex their arms at the elbow
- Examiner will instruct the patient to push against the
- examiner's resistance.

Upper Extremities-MS- Abduction

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Shoulder

Spinal nerve root innervations: C5, C6, Axillary nerve

- Ask the patient to raise their arm from the side to shoulder level
- The examiner presses down firmly on the patient's upper arm with the shoulder abducted
- Instruct the patient to abduct against resistance

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Wrist MS – Flexion/Extension

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Flexion

- Spinal nerve root innervations: C6, C7, Median Nerve
- Instruct the patient to make a fist
- The patient flexes their wrist down against examiner's resistance pushing it upward.

Extension

Spinal nerve root innervations: C6, C7, C8, radial nerve

- Instruct the patient to make a fist
- The patient should push up (extend their wrist) to resist as the examiner pushes down on the fist

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Finger MS- Adduction/Abduction

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Finger Adduction - MS

Spinal nerve root innervations: C7,C8, T1

- The patient is asked to squeeze two of the examiner's fingers as hard as they can and to not let go

Finger Abduction- MS

Spinal nerve root innervations: C8, T1, Ulnar nerve

- Position the patients' hand with the palm facing down and fingers spread.
- Instruct the patient to prevent you from moving any fingers as you try to force them together

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Opposition of the Thumb- MS

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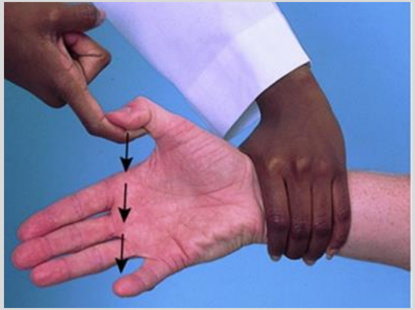
Spinal nerve root innervations:
C8, T1, median nerve

- Ask the patient to touch the tip of your finger with their thumb against your resistance



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Lower Extremities –MS-Hip Flexion/Extension

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Flexion

Spinal nerve root innervations: L2, L3, L4

- Examiner’s hand is on the anterior part of the patient’s mid thigh
- Patient is instructed to raise their leg up against the examiner’s resistance

Extension

Spinal nerve root innervations: S1

- Examiner’s hand is on the posterior part of the patient’s mid thigh
- Patient is instructed to push their thigh down against the examiners resistance

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Lower Extremities- MS- Hip Adduction/Abduction

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Adduction

Spinal nerve root innervations: L2, L3, L4

- Examiner places their hands on the inside of the patient's knees
- Patient is instructed to bring their legs together against the examiner's resistance

Abduction

Spinal nerve root innervations: L4, L5, S1

- Examiner places their hands on the outside of the patient's knees
- Patient is instructed to spread their legs apart against the examiner's resistance

Lower Extremities- MS Knee Flexion/Extension

Flexion

Spinal nerve root innervations: L5, S1, S2

- Patient is instructed to pull (flex) their legs against the examiner's resistance

Extension

Spinal nerve root innervations: L2, L3, L4

- Patient is instructed to push (extend) their legs against the examiner's resistance

Lower Extremities- MS- Ankle

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Dorsiflexion

Spinal nerve root innervations: L4, L5

- Patient is instructed to bring toes upward against resistance from the examiner

Plantar Flexion

Spinal nerve root innervations: S1

- Patient is instructed to push foot downward against resistance from the examiner

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Swartz 8th Edition, Chapter 21

Coordination

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Coordination for muscle movement requires:

- Motor system- muscle strength
- Vestibular system- balance, coordination of eye, head and body movements
- Sensory system- position sense
- Cerebellar system- integrates the information to produce normal rhythmic movement and steady posture

Assessment of Coordination

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Observe patient performing various movements:

- Rapid alternating movements

  Diadochokinesis

  Dysdiadochokinesis
- Point to point movements
- Gait and other related body movements
- Standing in specified ways

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Rapid Alternating Movements

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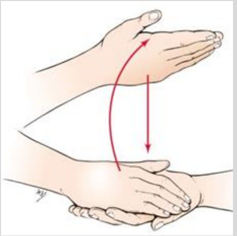
- **Arms (using hands)**

  Rapid alternation of pronation of one hand with supination on the other hand



- **Fingers**

  Touch the thumb to each finger quickly

- **Legs**

  Tap the floor or examiner's hand with the ball of the foot quickly

Observe for the speed, rhythm and smoothness of the movements

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Swartz, 7th Edition Chapter 18

Point to Point Movements- Arms/Legs

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Arms

Finger to Nose

- Patient alternates touching the Examiner’s index finger and then their own nose.
- Examiner changes the position of their finger.
- Observe for smoothness and accuracy
- Dysmetria- overshoot the mark initially and then reach it

Legs

Legs-Heel-Shin

- Patient places one heel on the opposite knee and will slide it down the shin of the other leg
- Observe for smoothness and accuracy

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Gait

- Uncoordinated gait can increase the risk of falls
- Ataxic gait- uncoordinated gait, unstable; seen in cerebellar disease, intoxication, loss of position sense

Bates 13th Edition, Chapter 24

Gait- Evaluation

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Walk across the room/down the hall

- Observe posture, balance, the swinging of the arms, and leg movements
- Normal- intact balance, arms swing symmetrically at the sides and turns are smooth
- Tandem Walking- walk in straight line using heel to toe

  May indicate ataxia not otherwise seen
- Walk on toes then heels
- Hop in place
- Shallow knee bend- on one leg then the other

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Reflexes

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- May be the earliest and most subtle indication of a neurological problem
- Involuntary response to a sensory stimulus
- Abnormalities are difficult to simulate
- Reflexes may be motor, sensory or autonomic
- Reflexes involve a stimulation of a receptor-ex: skin, mucous membranes, muscle, periosteum, retina, viscera

DeJong's The Neurologic Examination, 7th Edition, Chapter 37

Reflexes

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Types:

- Muscle Stretch Reflexes or Deep Tendon Reflexes (DTR)- protective function especially with standing and walking; involves specific spinal segments; monosynaptic reflex
- Cutaneous Reflexes-elicited by superficial skin stimulus and occurs in the same general area
- Primitive Reflexes- reflexes not normally present in adults

Bates, 13th Edition Chapter 24
DeJong's The Neurologic Examination, 7th Edition
Chapter 39; Harrison's 19th Edition Chapter 437

Reflexes

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Hyperreflexia (Hyperactive) Reflexes

- CNS lesions of the descending corticospinal tract
- Upper motor neuron findings of weakness, spasticity or positive Babinski sign

Hyporeflexia (Hypoactive) or Absent Reflexes

- Lesions of spinal nerve roots, spinal nerves, plexuses or peripheral nerves.
- Lower motor neuron disease- weakness, atrophy, fasciculations

Bates 13th Edition Chapter 24

Muscle Stretch Reflexes or Deep Tendon Reflexes

Scale for Grading Reflex: 0-4

4- Very Brisk, hyperactive, with clonus
(rhythmic oscillations between flexion and extension)

3- Brisker than average; possibly but not necessarily indicative of disease

2- Average; normal

1- Somewhat diminished; low normal

0- Reflex is absent

Document- 1+, 2+, 3+, 4+

Reflex Hammers

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Deep Tendon Reflex

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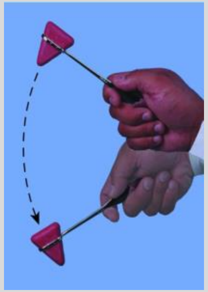
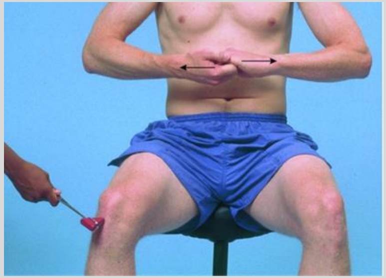
Procedure

- Patient should be encouraged to relax
- Reflex hammer should be held loosely between your thumb and index finger so that it swings freely in an arc within the limits of your palm and fingers
- Note speed, force and amplitude of the reflex response and grade it using the grading scale



Use techniques listed below for patients who have symmetrically diminished or absent reflexes

- For arm reflexes- ask patient to clench their teeth or squeeze their knees together
- For leg reflexes- ask patient to lock their fingers and pull against the other (Jendrassik's maneuver)



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Swartz, 7th Edition, Chapter 18

Deep Tendon Reflexes

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Biceps- Spinal nerve roots- C5,C6

- Examiner partially flexes the patient’s arm
- Strike the biceps tendon (alternative- examiner places their thumb or finger on the biceps tendon and then strike their finger)
- Assess bilaterally

Brachioradialis- Spinal nerve roots- C5,C6

- Patient’s hand is in a relaxed position-on lap, table or supported by the examiner’s arm.
- Examiner strikes the radius about 1 or 2 inches above the wrist with the reflex hammer
- Assess bilaterally

Bates, 13h Edition, Chapter 24

Deep Tendon Reflexes

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Triceps- Spinal nerve roots- C6,C7

- Examiner flexes the patient's arm at the elbow and supports the upper arm. Ask the patient to let their arm go limp.
- Strike the triceps tendon with the reflex hammer
- Assess bilaterally

Quadriceps (Patellar)-Spinal nerve roots L2, L3,L4

- Have patient sit on the edge of the table with legs flexed
- Strike the patellar tendon using the reflex hammer
- Assess bilaterally

Bates, 13th Edition Chapter 24

Deep Tendon Reflexes

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Achilles (Ankle)

Spinal nerve root (Primarily S1)

- Have patient remove shoes and socks
- Examiner dorsiflex the patient's foot at the ankle
- Achilles tendon is struck with the reflex hammer
- Assess bilaterally

Bates 13th Edition   Chapter 24, Hoppenfeld Chapter 8

Ankle Clonus

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- Clonus is tested when the reflexes appear to be hyperactive.
- Knee is supported in flexed position
- Dorsiflex and plantar flex the foot to relax it
- Sharply dorsiflex the foot
- Response- rhythmic oscillations
- Must be present in order to grade a

  stretch muscle reflex 4+

Cutaneous or Superficial Stimulation Reflexes

Elicited by superficial skin stimuli and occurs in the same general area

- Anal Reflex (Anocutaneous) S2,S3,S4
- Cremasteric Reflex- L1,L2
- Abdominal Reflex- T8- T12

Bates, 12th Edition, Chapter 17
DeJong's The Neurologic Examination, 7th Edition, Chapter 39
Swartz, 7th Edition, Chapter 18

Primitive Reflexes

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- Not normally present in adults
- Present in childhood
- May be associated with disease of the frontal lobe pathway
- Examples:
  - Grasp reflex
  - Palmomental response
  - Rooting reflex

Harrison's 19th Edition Chapter 437

Special Techniques

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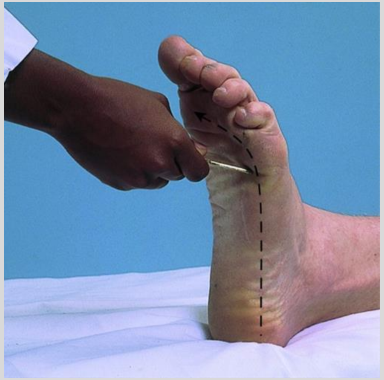
- Plantar Response

Examiner strokes the lateral aspect of the sole of the foot from the heel to the ball of the foot



Plantar Dorsiflexion-

Babinski Response- CNS lesion affecting the Corticospinal Tract

Plantar Flexion- Normal

Bates, 13th Edition, Chapter 24

Special Technique

Hoffmann's Test

Test for upper motor neuron lesion- Above T1

- Flick the nail of the middle finger
- If the thumb and 2nd finger flex, this is a positive Hoffman's Test. (See figure 2.70)
- The patient has neurological symptoms due to an upper motor neuron lesion and not to a radiculopathy or peripheral nerve lesion

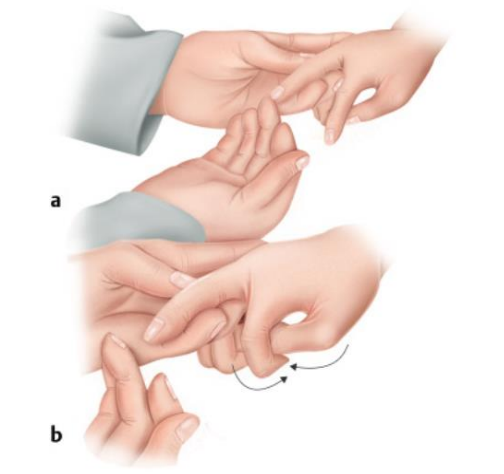


Fig. 2.70 (**a**) Hoffmann's test. (**b**) Positive Hoffmann's test with finger and thumb flexion after flicking the middle fingernail.

Special Techniques

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- Romberg Test
- Test of position sense
- Feet together with eyes open
- If there is no loss of balance continue
- Close both eyes for 30-60 seconds

Positive Romberg Test

- Loss of balance when eyes are closed

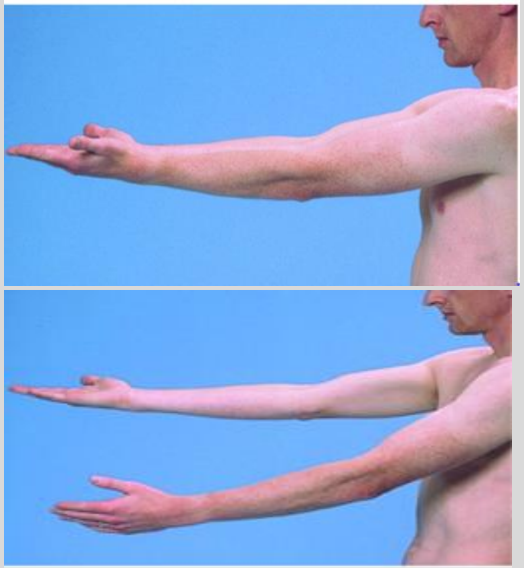
  (Dorsal Column disease)

Bates, 12th Edition Chapter 17 and Bates 13th Edition, Chapter 24

Special Techniques

- Pronator Drift Test

- Eyes closed and both arms straight forward with palm up for 20-30 seconds
- Maintain above position
- Examiner taps the arms briskly downward.



Normal response- arms return to original position

Pronator Drift- one forearm and palm turn inward and down (corticospinal tract lesion in contralateral hemisphere)

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Documentation of a Normal Neurological Examination

Neuro-

Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension

- CN I not tested CN II-XII grossly intact
- Motor strength is 5/5 in all extremities, good muscle bulk and tone
- Rapid alternating movements and finger to nose are intact
- Gait is normal, no evidence of ataxia
- Negative Romberg and Pronator Drift
- Deep Tendon Reflexes 2+ bilaterally, plantar reflexes downgoing or plantar flexion bilaterally

Bates, 13th Edition Chapter 24
Swartz, 7th Edition, Chapter 19

References

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- Bates Guide to Physical Examination and History Taking, 13th Edition, Lynn S. Bickley, MD, FACP, Chapter 13 (Ears and Nose) and (Chapter 24 (The Nervous System)
- DeJong's The Neurologic Examination, 7th Edition, William W. Campbell 2013; Chapter 37 (Introduction to Reflexes), Chapter 39 (Cutaneous Reflexes)
- Harrison's Principles of Internal Medicine, 19th Edition, Dennis L. Kasper, MD, Stephen L. Hauser, MD, J. Larry Jameson, MD, PhD, Anthony S. Fauci, MD, Dan L. Longo, MD and Joseph Loscalzo, MD, PhD 2015; Chapter 437 (Neurological Disorders)
- Neurologic Differential Diagnosis- A Case - Based Approach, Alan B. Ettinger, MD,MBA and Deborah M. Weisbrot, MD- 2014; Chapter 1 (Introduction)
- Physical Examination of the Spine and Extremities, Stanley Hoppenfeld, MD- 1976; Chapter 8 (Physical Examination of the Foot and Ankle

References

- Textbook of Physical Diagnosis History and Examination, 7th Edition, Mark H. Swartz, MD, FACP, Chapter 17 (The Musculoskeletal System), Chapter 18 (The Nervous System), Chapter 19 (Putting the Examination Together)

- Textbook of Physical Diagnosis History and Examination, 8th Edition, Mark H. Swartz, MD, FACP, Chapter 21 (The Nervous System)

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Preparation for Neuro Lab

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- Please be dressed in DPR lab attire
- Please bring a charged Ophthalmoscope, Visual Acuity card (Rosenbaum card), Penlight, Tuning Fork and Reflex Hammer
- Review Neurological DPR Skills Sheet as listed on the CPG

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Thank you

Lecture Feedback Form:

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